2ND SEMESTER LAB REPORT

TITLE : VERIFCATION OF OHM'S LAW



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BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

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Lab Engineer

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	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 3

Lab Title: VERIFCATION OF OHM'S LAW

Objective: 1. To verify voltage is equal to product current and resistance or verify ohm's Law

Student Name: **Muhammad Waqas -Muhammad Zeesha -Faiq Ali**

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 2/March/2025 Signature:

LABORATORY EXPERIMENT NO. 2

VERIFICATION OF OHM'S LAW

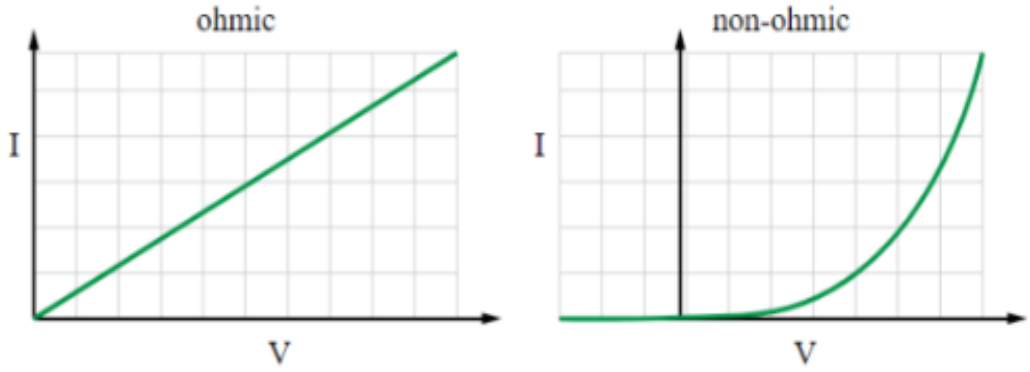
OBJECTIVE:

1. To verify Ohm's Law by calculating, and then by measuring voltage, current and resistance, and then comparing the calculated and measured results.
2. To verify Ohm's Law relationships by increasing one quantity while holding the second quantity constant, then measuring and calculating the effect on the third quantity.
3. To demonstrate the three power formulae in DC circuits.

Ohm's Law:

Ohm's Law is the basis of many electrical circuit calculations and is one of the most important theories you will learn: V=IR.
The current through a resistor is proportional to the voltage across it. The way in which we accomplish this is to measure the voltage across and current through a known
resistor for several different pair of values. We can then plot the data on a graph, and if the relationship is truly linear it should yield a straight line.
When graphing data such as those obtained in this experiment, either the x or y-axis can be chosen to display the voltage or current values. When the y-axis is chosen

I=V / R, V=IR and R= V / I



<<<<<<<<<<<<<<<<<<<<<(Lab Tasks)>>>>>>>>>>>>>>>>>>>>>

TASK 1: Perform the following steps to verify that

I = V / R.

========(a)========

Measure the resistance of the 6.8 kΩ (color-coded value) resistor with the DMM.



Orange-white-red-gold 38 X 10^2=3.8k Ω
DDM VALUE: 3.68 Ω

========(b)========

b. Connect the circuit as in the figure below.





c):Adjust the power supply voltage to 15.0 V using the DMM set on its DC voltage mode

==========(**d**)==========**.**

Measure and record the voltage across the R1 resistor) VR1= 5.08V



==========(**e**)==========



Measure and record the current through R1. (as shown in Figure).

(i): IR1 = <u>4.1584V</u> **(ii): P =VI = 4.1584*1.13** mili ampere**=4.6989 X 10^-3**



=========(f)========

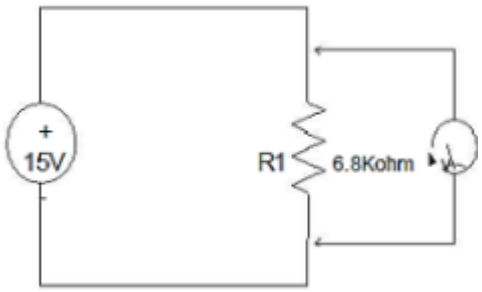
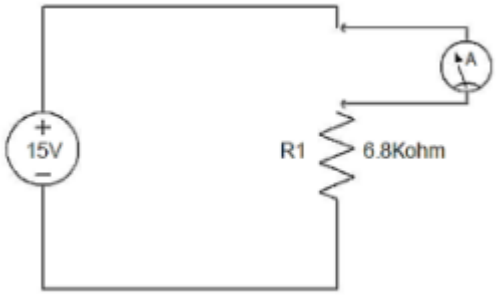
Using the measured values of voltage and resistance, calculate the current through R1

using Ohm's Law. Show calculations in the space below.

V=IR1 = 1.13mA X6.38K Ω =7.2094V I_{R1}=1.13mA

According to Ohm's Law:

I=V/R=7.209/3.68K Ω =1.1278mA **Hence Ohm's Law is verifeid**

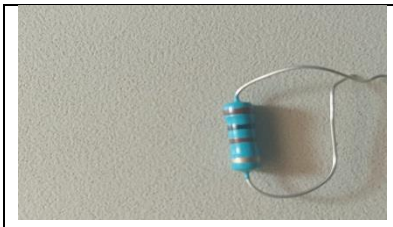
	
In this image resistor is connect with DMM in parallel to measure Voltage in resistor by providing voltage from power supply	In this image resistor is connect with DMM in seriesl to measure Current in resistor by providing voltage from power supply

==========(g)==========

What conclusion can you make from these procedures?

- Resistor is connect with DMM in parallel to measure Voltage
- Resistor is connect with DMM in seriesl to measure Current in resistor

TASK 2. Perform the following steps to verify that R =V / I



3.98K Ω

a. Connect the same circuit as in task 1, except change the value of resistance to 3.9 kΩ (color-coded value).

==========(b)==========.

Measure and record the voltage across R1 and the current flow through R1.

VR1 = 10.03V IR1 = 2.47mA

V=IR=(2.47mA)(3.9KΩ)=9.633V **Hence Ohm's Law is verified.**

==========(C)=========.

Calculate the resistance of R1 using the measured values of VR1 and IR1 with Ohm's Law.

Measured values are:

VR1 = 10.03V IR1 = 2.47mA

Show calculation. R1 = V_{R1}/I_{R1}=10.03V/2.47mA=4.06Ω

- Resistance from measured values=4.06K Ω
- Resistance from band values=3.9k Ω.

Difference=4.06K Ω-3.9k Ω.=0.160k Ω

=========(d)=========.

Remove R1 from the circuit. Measure and record the resistance of R1 using the multimeter.

R1 = 3.78KΩ P = I ^2 R = (2.47mA)^2 X 3.78K Ω =23 X 10^-3Watt

	
In this image resistor is connect with DMM in parallel to measure Voltage in resistor by providing voltage from power supply	In this image resistor is connect with DMM in seriesl to measure Current in resistor by providing voltage from power supply

Using the measured values of voltage and Current, calculate the resistance through R1.Using Ohm's Law.

V=10.03V I_{R1}=2.47mAmA

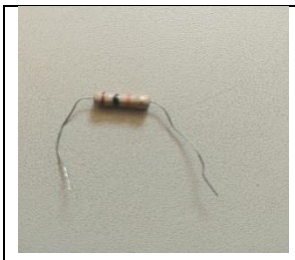
According to Ohm's Law:

R=V/I=7.209/2.47mA =4.06 Ω **Hence Ohm's Law is verif**

e. What conclusion can you make from these procedures?

- The purpose of this experiment is to verify Ohm's Law, which in words, simply says that the current through a resistor is proportional to the voltage across it.

TASK3. Perform the following procedures to verify that V = I ×R



10kΩ

a. Connect the circuit as in task 1 and 2, except change the resistor to 10 kΩ. (color-coded value)

========(b)========.

Measure the resistance of this resistor making sure that the resistor is removed from the circuit.

(Measured value) R1 = 9.37 **kΩ**

(Color Coded value) R1 = 10 **kΩ**

==========(c)==========

Reinsert the resistor into the circuit and measure the current through the resistor.

Voltage=15.03

I_R = **1.623mA (As resistance is heigh so current is minimun at 15.03V)**

=========(d)==========.

Calculate, using Ohm's law, the resistor voltage using the measured values of resistance and current. Show calculation below.

V = I × R = 1.623mA × 9.37KΩ =15.207V

============(e)============.

Measure the resistor voltage. Record

V_{R1} = 15.03V P= V^2/ R= 15.03^2/9.37K **Ω =24.108 X 10^-3 WATT**

==========(f.)==========

What conclusion can you make from these procedures?

To verify Ohm's Law (V = I × R) with R = 10 kΩ (10,000 Ω), first measure the voltage (V) across the resistor using a digital multimeter (DMM). Next, measure the current (I) by connecting the DMM in series with the resistor. Finally, calculate the resistance using R = V/I and confirm that it matches 10 kΩ, proving that voltage equals current times resistance

TASK4. Perform the following procedures to verify that

IF THE RESISTANCE IS HELD CONSTANT, INCREASING THE VOLTAGE WILL

INCREASE THE CURRENT.

a. Connect the circuit as in task 1,2 and 3 using a 4.7 kΩ resistor.

b. Set the voltage source to 2.0 V.

c. Measure the current through the resistor. Record this value in the following table.

d. Repeat this procedure from 2.0V to 30.0V. And note down the required values in the table given below.

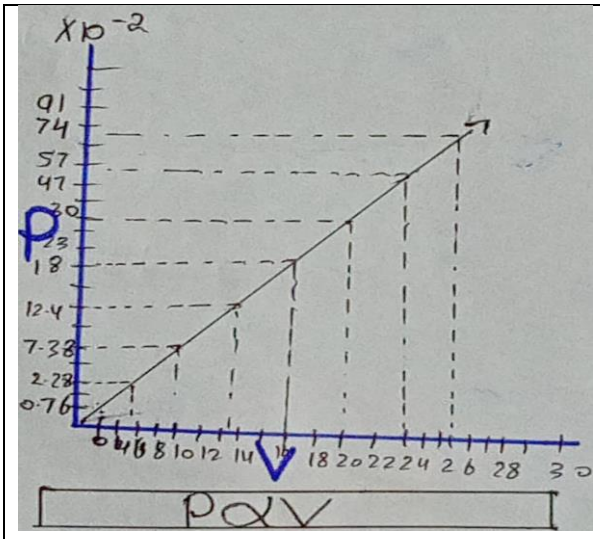
V_{R1}	I_{R1}	P_{R1}	R_1
0V	**0A**	0 WATT	**4.7 kΩ**
2V	0.38mA	0.76 x 10^-2 WATT	**4.7 kΩ**
4V	0.57mA	2.28 x 10^-2 WATT	**4.7 kΩ**
6V	1.23mA	7.38 x 10^-2 WATT	**4.7 kΩ**
8V	1.55mA	12.4 x 10^-2 WATT	**4.7 kΩ**
9V	2mA	18 x 10^-2 WATT	**4.7 kΩ**
10V	2.31mA	23.1 x 10^-2 WATT	**4.7 kΩ**
12V	2.54mA	30.24 x 10^-2 WATT	**4.7 kΩ**
14V	3.38mA	47.32 x 10^-2 WATT	**4.7 kΩ**
16V	3.57mA	57.12 x 10^-2 WATT	**4.7 k**
18V	4.12mA	74.16 x 10^-2 WATT	**4.7 kΩ**
20V	4.58mA	91.6 x 10^-2 WATT	**4.7 kΩ**
22V	5.22mA	114.8 x 10^-2 WATT	**4.7 kΩ**
24V	5.49mA	131.7 x 10^-2 WATT	**4.7 kΩ**
25V	6.22mA	155.5 x 10^-2 WATT	**4.7 kΩ**
26V	6.54mA	170.4 x 10^-2 WATT	**4.7 kΩ**
28V	6.6mA	184.8 x 10^-2 WATT	**4.7 kΩ**
30V	7.20mA	216.0 x 10^-2 WATT	**4.7 kΩ**

TASK5. Plot the following.

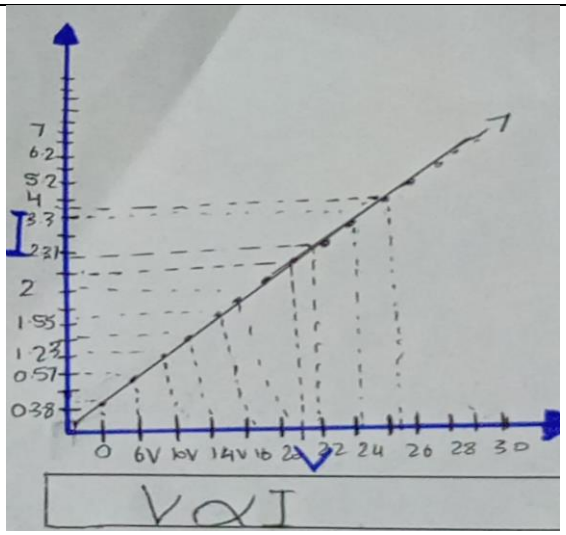
a) Draw relation between “V” & “I” on graph paper.

b) Draw relation between “P” & “V” on graph paper.

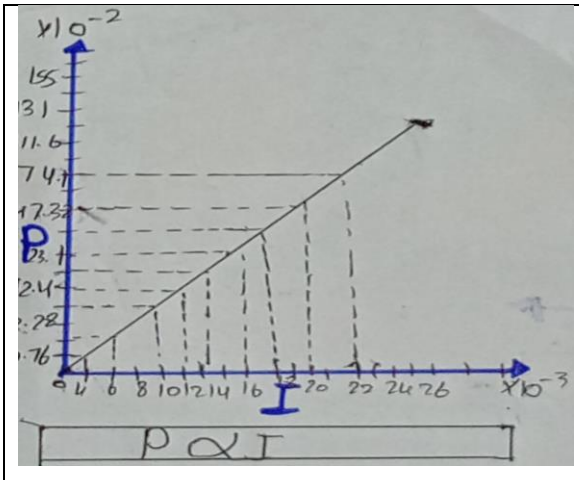
c) Draw relation between “P” & “I” on graph paper.



b): Power is directly peoportional to voltage



a): Voltage is directly peoportional to current



c**):Power is directly peoportional to current**

Conclusion:

In this experiment, we verified **Ohm's Law (I = V / R)** by measuring and calculating voltage, current, and resistance. First, we used a **digital multimeter (DMM)** to measure the resistance of a **6.8 kΩ resistor** and compared it with its color-coded value. Then, we connected the circuit and measured the voltage across the resistor and the current flowing through it. Using **Ohm's Law**, we calculated the expected current and found that it closely matched the measured value. This confirms that the relationship **I = V / R** holds true for an ohmic resistor, meaning the current increases when voltage increases while resistance remains constant.